

Description	Value				
0	session_id	2141	23	Iterative Imputation Iteration	None
1	Target	species	24	Numeric Imputer	mean
2	Target Type	Multiclass	25	Iterative Imputation Numeric Model	None
3	Label Encoded	Iris-setosa: 0, Iris-versicolor: 1, Iris-virgi...	26	Categorical Imputer	constant
4	Original Data	(150, 5)	27	Iterative Imputation Categorical Model	None
5	Missing Values	False	28	Unknown Categoricals Handling	least_frequent
6	Numeric Features	4	29	Normalize	False
7	Categorical Features	0	30	Normalize Method	None
8	Ordinal Features	False	31	Transformation	False
9	High Cardinality Features	False	32	Transformation Method	None
10	High Cardinality Method	None	33	PCA	False
11	Transformed Train Set	(104, 4)	34	PCA Method	None
12	Transformed Test Set	(46, 4)	35	PCA Components	None
13	Shuffle Train-Test	True	36	Ignore Low Variance	False
14	Stratify Train-Test	False	37	Combine Rare Levels	False
15	Fold Generator	StratifiedKFold	38	Rare Level Threshold	None
16	Fold Number	10	39	Numeric Binning	False
17	CPU Jobs	-1	40	Remove Outliers	False
18	Use GPU	False	41	Outliers Threshold	None
19	Log Experiment	False	42	Remove Multicollinearity	False
20	Experiment Name	clf-default-name	43	Multicollinearity Threshold	None
21	USI	6c50	44	Remove Perfect Collinearity	True
22	Imputation Type	simple	45	Clustering	False
			47	Polynomial Features	False
			48	Polynomial Degree	None
			49	Trigonometry Features	False
			50	Polynomial Threshold	None
			51	Group Features	False
			52	Feature Selection	False
			53	Feature Selection Method	classic
			54	Features Selection Threshold	None
			55	Feature Interaction	False
			56	Feature Ratio	False
			57	Interaction Threshold	None
			58	Fix Imbalance	False
			59	Fix Imbalance Method	SMOTE

Figure 3: PyCaret environment setup result

and time series analysis with PyCaret.

Classification model building with PyCaret

This article explains classification model building with PyCaret by taking the Iris data set from PyCaret's data repository.

We will use the Google Colab environment to make things simple and follow the steps mentioned below.

Step 1: First, install PyCaret by giving the following command:

```
pip install pycaret
```

Step 2: Next, load the data set, as shown in Figure 1:

```
from pycaret.datasets import get_data
dataset = get_data('iris')
(or)
import pandas as pd
dataset = pd.read_csv('/path_to_data/
file.csv')
```

Step 3: Now set up the PyCaret environment, as shown in Figure 2:

```
from pycaret.classification import *
clf1 = setup
(data=dataset, target = 'species')
```

For any type of model building with PyCaret, the environment setup is the most important step. By default, *setup()* function takes the *data*: Pandas DataFrame and *target*, which points to the class label variable in

the data set. The result of the setup function is shown in Figure 3. The setup function, by default, splits 70 per cent of the data as train set and 30 per cent as test set, and does data preprocessing as shown in Figure 3.

Step 4: Next, find the best model, as shown in Figure 4:

```
best = compare_models()
```

best = compare_models()									
	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
lda	Linear Discriminant Analysis	0.9818	0.9951	0.9806	0.9818	0.9818	0.9725	0.9725	0.017
qda	Quadratic Discriminant Analysis	0.9718	0.9976	0.9722	0.9743	0.9718	0.9576	0.9589	0.018
gbc	Gradient Boosting Classifier	0.9718	0.9942	0.9722	0.9784	0.9714	0.9576	0.9611	0.198
lr	Logistic Regression	0.9618	0.9951	0.9611	0.9704	0.9610	0.9422	0.9471	0.326
dt	Decision Tree Classifier	0.9527	0.9649	0.9528	0.9636	0.9519	0.9286	0.9346	0.015
lightgbm	Light Gradient Boosting Machine	0.9527	0.9976	0.9500	0.9590	0.9519	0.9283	0.9321	0.072
ada	Ada Boost Classifier	0.9518	0.9815	0.9472	0.9646	0.9480	0.9260	0.9345	0.093
knn	K Neighbors Classifier	0.9418	0.9837	0.9417	0.9549	0.9405	0.9119	0.9194	0.116
rf	Random Forest Classifier	0.9418	0.9920	0.9417	0.9576	0.9384	0.9114	0.9211	0.468
et	Extra Trees Classifier	0.9327	0.9951	0.9306	0.9462	0.9293	0.8975	0.9061	0.435
nb	Naive Bayes	0.9236	0.9896	0.9222	0.9383	0.9198	0.8843	0.8937	0.016
ridge	Ridge Classifier	0.8673	0.0000	0.8667	0.8793	0.8647	0.8001	0.8082	0.017
svm	SVM - Linear Kernel	0.7855	0.0000	0.8056	0.8055	0.7401	0.6850	0.7382	0.064
dummy	Dummy Classifier	0.3855	0.5000	0.3333	0.1489	0.2147	0.0000	0.0000	0.013

Figure 4: Finding the best model